

Andrew Scouten

andrewscouten@gmail.com , (325) 513-3296 , San Marcos, TX 78666

[in andrewscouten](#) , [✉ andrewscouten](#) , orcid.org/0009-0004-6418-7158

SUMMARY

Machine Learning engineer and researcher with experience building end-to-end data pipelines and training modern ML models for real-world applications. Contributed to published research in infrastructure-scale pavement analysis (TxDOT) and ISS-derived corrosion studies (NASA/SpaceX-21). Led multi-modal ML systems and cross-team projects, focusing on scalable, applied solutions.

EDUCATION

Texas State University

Master of Science in Computer Science

December 2026

Bachelor of Science in Computer Science

May 2023

EXPERIENCE

Graduate Research Assistant

May 2023 - Dec. 2025

Texas State University

San Marcos, TX

- Conducted research for the [Texas Department of Transportation \(Project 0-7150: AI for Pavement Condition Assessment from 2D/3D Surface Images\)](#).
- Built and maintained data pipelines for large scale pavement imaging datasets; designed, trained, and evaluated CNN and transformer-based models for crack detection, leading to a conference poster and peer-reviewed publication.

Undergraduate Researcher

Jan. 2023 - Dec. 2024

Texas State University

San Marcos, TX

- Contributed to [NASA-backed SpaceX-21 research on bacterial adhesion and corrosion](#) using International Space Station derived samples.
- Developed data organization and quality-control pipelines; applied classical machine learning and deep learning methods for corrosion segmentation and labeling, resulting in a peer-reviewed publication.

Java Developer

Jan. 2021 - Jan. 2024

LurgCraft, Minecraft Network

Remote

- Sole Java developer of [LurgCraft](#), engineered cross-server chat synchronization and implemented game-play features across multiple Minecraft versions and modding platforms.
- Delivered 20 custom projects and 46 enhancements to existing code-bases, strengthening ability to rapidly understand and extend legacy systems.

PROJECTS

Hackathons

2025, 2026

- Federated Learning (2026)*: Coordinated a team of 15 contributors across three sub-projects, guiding architecture and development. Co-developed [OncoLearn](#), a multi-modal ML toolkit for cancer genomics and biomarker discovery.
- Knowledge Graph (2025)*: Co-developed [ClassiGraph](#), a Graph Neural Network for multi-modal cancer subtype classification.

TXST Center for Analytics and Data Science (CADS)

2023, 2024

Texas State University

San Marcos, Texas

- Summer Workshop (2023)*: Collaborated with 3–5 non-CS researchers to apply AI methods to domain-specific problems over an intensive 3-day workshop.
- Annual Analytics Showcase (2023, 2024)*: Presented TxDOT AI research to non-technical audiences, improving ability to communicate complex concepts clearly.

SKILLS

Technical Skills: C, C++, Python, R, Java, JavaScript, CMake, Anaconda / UV, Maven / Gradle, Renv, Node.js, PyTorch, CUDA, Expo, React, React Native, SQL, PostgreSQL, MongoDB, Linux Systems, Git

Languages: German (Conversational, A2/B1)